

and software used will be Indian.

So what is little Maya going to be when she grows up? "There are a lot of applications being looked at—surveying the seabed, for under-sea mapping. If we get the right sensors, we can examine the topology of the seabed. Currently, we only use satellites that can only study the surface of the sea." The main task is to increase the ability to collect data in innovative ways. "I believe that if we can get the technology in place, we can have a family of them—small ones, big ones, ones that can dive, that can go horizontally, some that can explore by themselves," sums up de Sa. The battery technology restricts the scientists from keeping Maya underwater long enough to imprint the scenic landscape on its electrodes. In the future, de Sa hopes to use fuel technology that will increase its endurance since weight and power are the crucial factors that will decide its staying capacity.

We have some aspirants for the medical field too and misleadingly, one of them is called the da Vinci. The da Vinci

showdown—and the teams that won top honours included Jaws from Veer Mata Jeejabai Technical Institute (VJT), Mumbai, who came up on top at Level 1, The Titans from IIT Mumbai who triumphed at Level 2, and AI from IIT Kharagpur who won the Level 3 competition. At the Level 2 finals, it was a really close fight between The Titans and their adversary, Balbeer Pasha (also from IIT Mumbai) with both bots playing very hard to win. The winners got to take home cash prizes and Lego Mindstorm kits.



The Ping-Pong Titans from IIT-B, winners of the Level 2 competition

Two students from the Dortmund University, Germany put up a game of robot-soccer with six Sony Aibos. Each Aibo responded to the movement of the ball, and the concept of team play. And what's a mutt to do after winning? Why dance, of course—six puppies danced to their victory, in sync and step with each other!

console employed at the Escorts Heart Institute and Research Centre in Delhi is not known for its bedside manner. What it is known for, however, is aiding doctors during critical heart surgeries. The da Vinci bears close resemblance to its video gaming cousins and is accompanied by a surgical cart with three arms jutting out. The arms need to be assisted to the incisions made on the patient and one of them is fitted with an endoscope that transmits images of the patient's organ onto a colour monitor. The da Vinci recently aided a successful bypass heart surgery. The Escort Centre is currently one of the handful of hospitals to use robots in surgery. It needs to be pointed out here that robots are only used to assist qualified doctors; they do not perform the surgery by themselves.

The robot brings a lot of advantages to the operation table—it does away with the foot-long incisions needed in conventional open-heart surgery by using a smaller incision and displays the heart as a three-dimensional image, magnified 10 times its natural size to give clarity. It also takes over the function of artificially pumping the blood while the operation is in progress. But its special skill lies in easing performance anxiety—da Vinci deals with surgeon fatigue and hand tremors by providing a stand for the hand and steadying it. However, it will be sometime before da Vinci finds its way into more hearts. Today, the robotic heart surgery costs around Rs 2 lakh and the da Vinci costs Rs 7 crore to import. Industry experts expect the costs to come down in the next five years when the components and technology involved will be cheaper. It will also be cheaper for the patient, since the recuperation time spent in the hospital can get as short as three days.

In the real world

All the budding technologies have been late for their date with the subcontinent compared to their counterparts in Europe, the United States and Japan. These nations have been using robots for some time now. The primary hindrance is that robots have traditionally been looked upon as replace-

ments for human labour. "The main reason for the arrested growth of robotics in India," explains Amarnath, "is that most western countries use robots as an alternative for their expensive human labour. Another reason is that the components used to make these robots were not available in India till late."

So what makes them ready for the tropical subcontinental weather now? "The key," says Amarnath, "is to adapt robots to our economic scenario. We cannot use robots to reduce human labour.



Prof Subir Saha
ASSOCIATE PROFESSOR
IIT, DELHI

A published statistic shows that 3,50,000 robots account for less than 3 per cent unemployment in Japan

Instead, we must use it as a tool and to increase productivity and precision." Saha agrees: "We must use robotics since we are competing with countries such as the United States, China, Russia and Japan. It has been proved that precision cannot be achieved by manual labour."

Robots can also be used to increase productivity. In an industry such as carpet cleaning, a worker is paid by the number of carpets he or she cleans. Here, an automated tool increases the efficiency and does not replace the worker. Many industry experts feel that we should study the robots being used abroad and adapt and produce them indigenously. This will lower their costs, making them more accessible to companies.

The answer to the question that will judgement day ever be brought upon by machines, depends on their intelligence. The intelligence of a robot is measured by how autonomous it is. The definition of robots itself establishes that it has to be programmed to perform a task. Hence, the intelligence behind it will always be human. How well it performs the task and how easily it overcomes the obstacles it faces, will determine the intelligence of the machine. So while we may not yet have robots using us as batteries to power themselves, there is a possibility of them cleaning our carpets. 

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